HR Analytics: Employee Attrition Analysis

A complete data pipeline using Python, SQL, and Power BI to understand workforce trends and reduce employee turnover.

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Project Overview

01

Data Cleaning

Python-based preparation and quality checks

02

Database Integration

MySQL storage with SQLAlchemy

03

SQL Analysis

Business intelligence queries

04

Dashboard Reporting

Power BI visualization

This pipeline transforms raw HR data into actionable insights for data-driven retention strategies.

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Data Preparation in Python

Initial Exploration

- Dataset structure analysis
- Summary statistics generation
- Data type verification

Data Cleaning Steps

- Missing value imputation
- Column standardization
- Duplicate record removal

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Key Business Questions

Workforce Diversity

Employee distribution by gender

Income Analysis

High-earners and salary patterns

Attrition Patterns

Department and role analysis

Work-Life Balance

Compensation correlations

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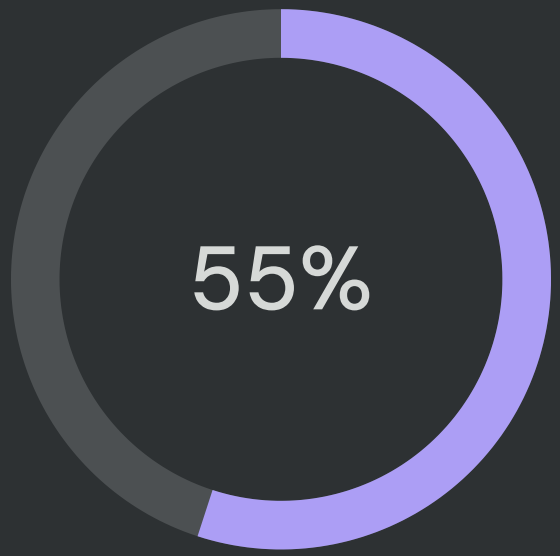
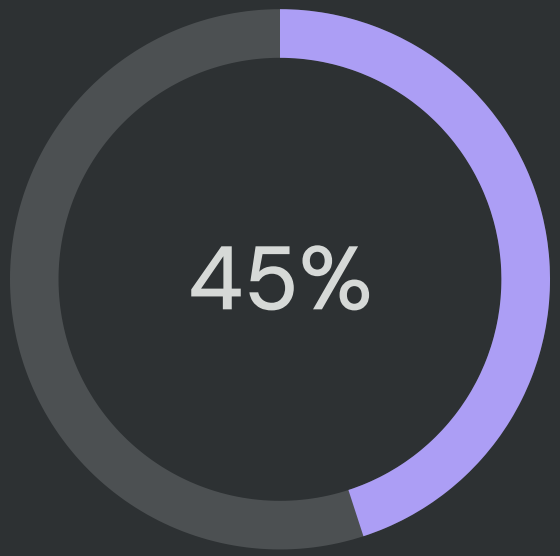
Employee Distribution by Gender



Male Employees

Majority workforce segment



Female Employees

Growing representation

Understanding gender distribution helps organizations assess diversity and inclusion initiatives.

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High-Income Employee Analysis

15%

Above Average

Employees earning above company median

$5,432

Average Monthly

Company-wide income baseline

$12,5...

Top Earner

Highest monthly income recorded

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Attrition Analysis Findings

High-Risk Departments

Department-level attrition patterns identified through SQL analysis

Top 5 At-Risk Roles

Job roles with highest turnover rates requiring immediate attention

Tenure Impact

YearsAtCompany shows correlation with reduced attrition probability

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Work-Life Balance Insights

Key Findings

- Average monthly income varies significantly across work-life balance levels
- Lower scores correlate with higher turnover risk
- Compensation patterns reveal retention opportunities

Impact Areas

Work-life balance scores directly influence employee satisfaction and retention rates.

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Power BI Dashboard

Interactive visualizations present real-time insights for HR decision-making.

Executive Summary

High-level metrics and KPIs

Attrition Trends

Historical patterns and forecasts

Compensation Analysis

Income distribution insights

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Strategic Recommendations

High-Attrition Departments

Investigate root causes: workload, management, growth opportunities

Work-Life Programs

Prioritize flexible policies and wellness initiatives

Salary Benchmarking

Review competitive compensation for key roles

Manager Engagement

Promote regular check-ins and leadership training

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